Session 3: Bayesian State Space Models

Modern Macroeconometrics

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Overview of Session 3

- **Motivation:** Why state space models in macroeconometrics?
- **The Gaussian linear state space model:** A general framework
- **Main models:** Unobserved components model (UCM) and TVP regression
- **Non-centered parameterisation:** Time variation viewed as model selection
- **Stochastic volatility (SV):** Heteroskedasticity via time-varying error variances
- **Goal:** A large TVP regression with SV and shrinkage priors that let the data decide which coefficients actually move
- **Final discussion (if time allows):** Static versus dynamic sparsity

Introduction and motivation

Motivation for more flexible models

Limits of constant parameter models

- Standard regression or VAR models assume **constant parameters** throughout the sample
- Overwhelming evidence that macro relationships are **time varying**, and when formally tested (e.g., Stock and Watson 1996, *JBES*), stability is often rejected:
  - Effectiveness of monetary policy transmission changes over time (Primiceri 2005, *ReStud*)
  - Inflation dynamics: Phillips curve flattening, anchoring of expectations, changes in inflation trends (Stock and Watson 2007, *JMCB*; Cogley et al. 2010, *AEJ:Macro*; Inoue et al. 2025, *ET*)
- **State space models** provide a natural framework

Key state space models in Bayesian macroeconometrics

- State space methods are used for a wide variety of time series problems
- They are important in their own right in economics (e.g., trend-cycle decompositions, structural time series models, dealing with missing observations, etc.)
- TVP regressions and SV models are both state space models
- DSGE models are state space models (DYNARE is a popular Bayesian toolbox for estimation)
- A key advantage: Well-developed MCMC algorithms exist for Bayesian inference in state space models

Our goal for the morning session

- Build up step by step towards a **large TVP regression with SV and shrinkage**:

$$
\begin{aligned}
y_t &= x_t' \boldsymbol{\beta}_t + \exp\left(\frac{h_t}{2}\right)\varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0,1) \\
\boldsymbol{\beta}_t &= \boldsymbol{\beta}_{t-1} + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega}) \\
h_t &= \mu_h + \phi_h(h_{t-1} - \mu_h) + \eta_{h,t}, \quad \eta_{h,t} \sim \mathcal{N}(0, \omega_h)
\end{aligned}
$$

- Three components we need: 1. **TVPs** ($\boldsymbol{\beta}_t$), 2. **shrinkage priors on $\boldsymbol{\Omega}$** (let the data decide which β_{jt} actually vary over time), 3. **SV** (h_t)
- Each component has a modular sampling step; **power of Bayesian econometrics** is that these combine naturally into a single Gibbs sampler Appendix

The Bayesian perspective: likelihood and prior

- Bayesian inference requires two ingredients: A **likelihood** and a **prior**
- **Likelihood** is defined by so-called **observation equation**

$$y_t \mid \beta_t, h_t \sim \mathcal{N}\left(x_t'\beta_t,\ \exp(h_t)\right)$$

- **Prior** is defined by the evolution equations (so-called **state equations**)
  - Beliefs about **how states change over time**; this is a modelling choice; but some flexibility via **adaptive shrinkage**
  - $\beta_t = \beta_{t-1} + \eta_t$: Random walk implies parameters drift **gradually and permanently**
  - $h_t = \mu_h + \phi_h(h_{t-1} - \mu_h) + \eta_{h,t}$: AR(1) implies volatility is **persistent but mean reverting**
  - Other choices (e.g., regime-switching) reflect different beliefs on nature of instability

Gaussian linear state space model

The Gaussian linear state space model

- The **measurement equation** links observables to latent states:

$$\boldsymbol{y}_t = \boldsymbol{Z}_t \boldsymbol{\beta}_t + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma}_t)$$

- The **state equation** describes the evolution of the latent states:

$$\boldsymbol{\beta}_t = \boldsymbol{T}_t \boldsymbol{\beta}_{t-1} + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Omega}_t)$$

- **Key notation**:
  - $\boldsymbol{y}_t$: $M \times 1$ vector of observations; $\boldsymbol{Z}_t$: $M \times K$ known design matrix
  - $\boldsymbol{\beta}_t$: $K \times 1$ vector of unobserved **states** (e.g., time-varying coefficients)
  - $\boldsymbol{T}_t$: $K \times K$ transition matrix (usually $= \boldsymbol{I}_K$ for random walk states)
  - $\boldsymbol{\varepsilon}_t$ and $\boldsymbol{\eta}_s$ are independent for all t, s

How to obtain TVP regression?

- Setting $\boldsymbol{Z}_t = \boldsymbol{x}_t'$, $\boldsymbol{T}_t = \boldsymbol{I}_K$, $\boldsymbol{\Sigma}_t = \sigma^2$, and $\boldsymbol{\Omega}_t = \boldsymbol{\Omega}$ gives:

$$y_t = \boldsymbol{x}_t'\boldsymbol{\beta}_t + \varepsilon_t, \qquad \boldsymbol{\beta}_t = \boldsymbol{\beta}_{t-1} + \boldsymbol{\eta}_{\beta,t}$$

  a **TVP regression** with coefficients following random walks
- Allowing $\sigma_t^2 = \exp(h_t)$ introduces **SV**
- The normal linear state space model is very useful
- Well-developed MCMC algorithms; in particular forward filtering backward sampling (FFBS) ▸ Appendix
- But some models have y_t as a **nonlinear** function of the states (e.g., SV mentioned above; workarounds needed!)

Zoom into state equation

- The role of $\mathbf{\Omega}$:

$$\boldsymbol{\beta}_t = \boldsymbol{\beta}_{t-1} + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{\Omega})$$

- $\mathbf{\Omega} = \text{diag}(\omega_1, \ldots, \omega_K)$ controls **how much each coefficient is allowed to drift**
  - $\omega_j = 0$: coefficient j is constant throughout the sample
  - $\omega_j > 0$: coefficient j is time varying
- **Note:** $\boldsymbol{\beta}_0$ (so-called **initial state**) enters equation for $\boldsymbol{\beta}_1$ and therefore requires its own prior

Special case of TVP regression

- Set $\boldsymbol{x}_t = 1$ and $\boldsymbol{\beta}_t = \alpha_t$, the TVP regression reduces to an **unobserved components model (UCM)**

$$
\begin{aligned}
y_t &= \alpha_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2) \\
\alpha_t &= \alpha_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \omega)
\end{aligned}
$$

- α_t is an unobserved **random walk trend**; $\boldsymbol{\Omega} = \omega$ is now a scalar
- **Signal-to-noise ratio** $q = \omega/\sigma^2$ governs how quickly the trend moves:
  - $q \to 0$: Trend is nearly constant ($\alpha_t = \alpha$)
  - $q \to \infty$: Trend perfectly fits the data ($y_t = \alpha_t$)

Non-centered parameterisation illustrated with UCM

Frühwirth-Schnatter and Wagner (2010, *JoE*)

Non-centered parameterisation

- To discuss shrinkage in a TVP regression, we first need to discuss the non-centered parameterisation
- Assume $y_t = \alpha_t + \varepsilon_t$, with $\alpha_t = \alpha_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \omega)$
- We can write $\alpha_t = \alpha_0 + \sqrt{\omega}\tilde{\alpha}_t$, with $\tilde{\alpha}_0 = 0$ and

$$\tilde{\alpha}_t = \tilde{\alpha}_{t-1} + \tilde{\eta}_t, \quad \tilde{\eta}_t \sim \mathcal{N}(0, 1)$$

- Decompose the time-varying states into a **constant part** (α_0) and **deviations from that constant** ($\sqrt{\omega}\tilde{\alpha}_t$)

Non-centered parameterisation (cont.)

- This allows us to rewrite the observation equation as:

$$y_t = \underbrace{\alpha_0 + \sqrt{\omega}\tilde{\alpha}_t}_{\alpha_t} + \varepsilon_t$$

- Conditional on $\tilde{\alpha}_t$, we can estimate both α_0 and $\sqrt{\omega}$ in a standard regression model by defining $\boldsymbol{x}_t = (1, \tilde{\alpha}_t)'$ and $\boldsymbol{\beta} = (\alpha_0, \sqrt{\omega})'$
- Presence of time variation becomes a **model selection problem**!
  - If $\sqrt{\omega} = 0$, we end up with a constant parameter regression
  - $\omega > 0$ implies α_t is time varying
  - Non-centered parameterisation now allows to use proper shrinkage priors on $\sqrt{\omega}$ (see Session 2)

Stochastic volatility (SV)

Nonlinear state space models

- The normal linear state space model is very useful for empirical macroeconomists (e.g., trend-cycle decompositions, homoskedastic TVP regressions, and linearised DSGE models)
- Some models have $\boldsymbol{y}_t$ as a **nonlinear** function of the states (e.g., DSGE models that have not been linearised)
- Standard Bayesian algorithms cannot be used
- There is a growing set of Bayesian tools for nonlinear state space models (e.g., particle filter)
- Here we focus on **stochastic volatility**: Practically most important nonlinear state space model for macroeconomics and finance

Linearising the measurement equation

- The SV model has measurement equation $y_t = \exp\left(\frac{h_t}{2}\right)\varepsilon_t$, with $\varepsilon_t \sim \mathcal{N}(0,1)$
- Square and take the log of the measurement equation to obtain:

$$y_t^* = h_t + \varepsilon_t^*$$

  with $y_t^* = \ln(y_t^2)$ and $\varepsilon_t^* = \ln(\varepsilon_t^2)$
- The measurement equation is now **linear** in h_t
- The error $\varepsilon_t^* = \ln(\varepsilon_t^2)$ is **no longer Gaussian**: $\varepsilon_t^* \sim \ln \chi_1^2$
- Linearising results in a linear state space model but with **non-Gaussian errors**
- **Key idea:** Approximate well-known $\varepsilon_t^* \sim \ln \chi_1^2$ distribution with a **mixture of Normal distributions**

Mixture of Normals approximation

- Mixtures of Normals are very flexible and widely used to approximate unknown or inconvenient distributions
- Once approximated, we can use all techniques from the Gaussian linear state space model (FFBS)
- Kim, Shephard and Chib (1998, *ReStud*) use a **7-component mixture**:

$$\ln \chi_1^2 \approx \sum_{i=1}^{7} q_i\, f_N(\varepsilon_t^* \mid m_i, v_i^2)$$

- Since $\varepsilon_t^* \sim \ln \chi_1^2$ involves no unknown parameters: q_i, m_i, and v_i^2 for $i = 1, \ldots, 7$ are fixed numbers (see Table 4 of Kim, Shephard and Chib, 1998)
- Omori, Chib, Shephard, and Nakajima (2007, *JoE*) use 10 components

Unobserved components stochastic volatility (UC-SV) model

The UC-SV model: Set-up

- Adds SV to a stochastic trend (random walk intercept):

$$y_t = \alpha_t + \exp\left(\frac{h_t}{2}\right)\varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0,1)$$

$$\alpha_t = \alpha_{t-1} + \eta_{\alpha,t}, \quad \eta_{\alpha,t} \sim \mathcal{N}(0,\omega_\alpha)$$

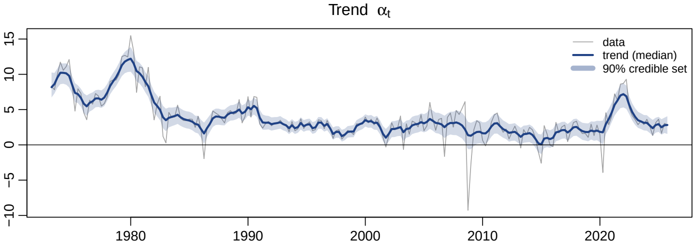
$$h_t = \mu_h + \phi_h\,(h_{t-1} - \mu_h) + \eta_{h,t}, \quad \eta_{h,t} \sim \mathcal{N}(0,\omega_h)$$

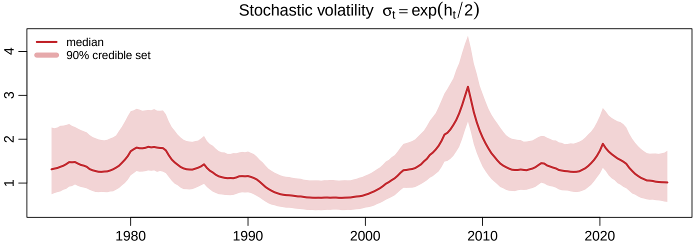
- Model often performs well for inflation; α_t is interpreted as **trend inflation**

The UC-SV model: An illustration with inflation

- Estimates of **trend inflation** α_t with 90% posterior credible intervals
- Estimates of the **time-varying volatility** process $\exp(h_t/2)$ with 90% posterior credible intervals
- The results will show how inflation trend and uncertainty have changed in the US in the last 5 decades

The UC-SV model: An illustration with inflation





Goal: TVP regression with SV and shrinkage

A TVP regression with SV

- Our **full model**: TVPs ($\boldsymbol{\beta}_t$), SV (h_t) and shrinkage on $\boldsymbol{\Omega}$:

$$y_t = \boldsymbol{x}_t'\boldsymbol{\beta}_t + \exp\left(\frac{h_t}{2}\right)\varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0,1)$$

$$\boldsymbol{\beta}_t = \boldsymbol{\beta}_{t-1} + \boldsymbol{\eta}_{\beta,t}, \quad \boldsymbol{\eta}_{\beta,t} \sim \mathcal{N}(\boldsymbol{0},\boldsymbol{\Omega})$$

$$h_t = \mu_h + \phi_h(h_{t-1} - \mu_h) + \eta_{h,t}, \quad \eta_{h,t} \sim \mathcal{N}(0,\omega_h)$$

- MCMC is **modular**, with each block conditioning on the others ▸ Appendix
  1. Draw $\boldsymbol{\beta}^{(1:T)}$ via FFBS, conditioning on $\boldsymbol{\Omega}$ and $\boldsymbol{h}^{(1:T)}$
  2. Draw $\boldsymbol{\Omega}$ (or individual ω_j's with shrinkage) conditioning on $\boldsymbol{\beta}^{(1:T)}$
  3. Draw $\boldsymbol{h}^{(1:T)}$ via the Kim, Shephard and Chib (1998) algorithm, conditioning on residuals
  4. Draw SV parameters $(\mu_h, \phi_h, \omega_h)$ from standard conditionals

The curse of dimensionality in TVP regressions

- In large predictive regressions (e.g., forecasting inflation with $K = 100$ predictors), we must estimate TK time-varying coefficients from only T observations (by construction $TK \gg T$)
- The **random walk state equation** is a first regularisation device, tying $\boldsymbol{\beta}_t$ to $\boldsymbol{\beta}_{t-1}$
- But even then, **which β_{jt} actually vary** is unknown a priori
- Common approaches to cope with this:
  - **Diagonal Ω**: coefficients evolve independently (most parsimonious)
  - **Non-centered parameterisation** + global–local shrinkage on $\sqrt{\omega_j}$: data-driven, tuning-free

Non-centered parameterisation of TVP regression

- Generalise the UCM to K regressors/coefficients:

$$\boldsymbol{\beta}_t = \boldsymbol{\beta}_0 + \sqrt{\boldsymbol{\Omega}}\,\tilde{\boldsymbol{\beta}}_t, \qquad \tilde{\boldsymbol{\beta}}_t = \tilde{\boldsymbol{\beta}}_{t-1} + \tilde{\boldsymbol{\eta}}_t$$

with $\sqrt{\boldsymbol{\Omega}} = \text{diag}(\sqrt{\omega_1}, \ldots, \sqrt{\omega_K})$, $\tilde{\boldsymbol{\eta}}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}_K)$, $\tilde{\boldsymbol{\beta}}_0 = \mathbf{0}$

  - $\boldsymbol{\beta}_0$: the **constant coefficient part**
  - $\sqrt{\omega_j}$: the **square root of the state innovation variance** (time-variation of coefficient j)
  - $\tilde{\boldsymbol{\beta}}_t$: the **normalised states** (unit-variance random walk)

- Substituting into the observation equation:

$$y_t = \boldsymbol{x}_t'\boldsymbol{\beta}_0 + \sum_{j=1}^{K} x_{jt}\,\sqrt{\omega_j}\,\tilde{\beta}_{jt} + \exp\left(\frac{h_t}{2}\right)\varepsilon_t$$

Global-local shrinkage in TVP regressions

- Conditional on $\tilde{\boldsymbol{\beta}}_t$, both $\boldsymbol{\beta}_0$ and the scales $\sqrt{\omega_j}$ are ordinary regression coefficients
- Use the **horseshoe** (global-local) prior from Session 2 on $\boldsymbol{\beta}_0$ and each $\sqrt{\omega_j}$:

$$\beta_{0,j} \mid \phi_{\beta,j}, \lambda_\beta \sim \mathcal{N}\left(0, \phi_{\beta,j}^2 \lambda_\beta^2\right),$$

$$\sqrt{\omega_j} \mid \phi_{\omega,j}, \lambda_\omega \sim \mathcal{N}\left(0, \phi_{\omega,j}^2 \lambda_\omega^2\right)$$

  with locals $\phi_{\beta,j}, \phi_{\omega,j} \sim C^+(0,1)$ and globals $\lambda_\beta, \lambda_\omega \sim C^+(0,1)$
  - $\lambda_\beta, \lambda_\omega$ (**global**): Overall shrinkage
  - $\phi_{\beta,j}, \phi_{\omega,j}$ (**local**): Let an individual coefficient escape
- Incorporate SV as before

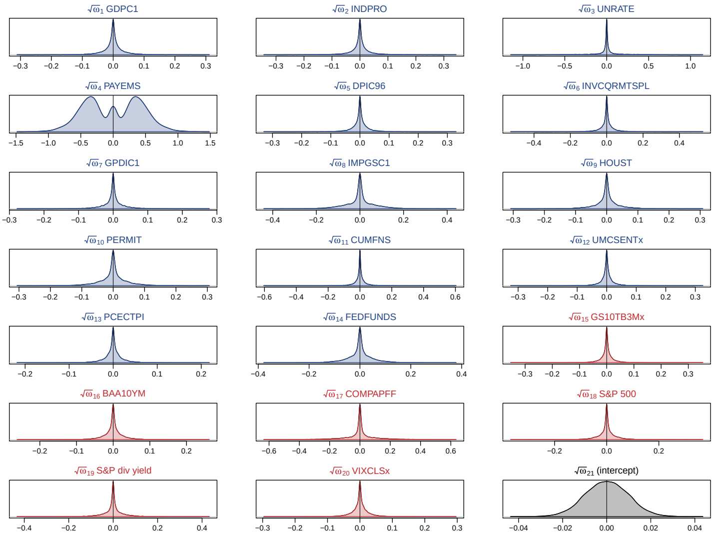
Application: Growth at risk with TVPs and SV

Plagborg-Møller et al. (2020, *Brookings Papers*) from Session 2

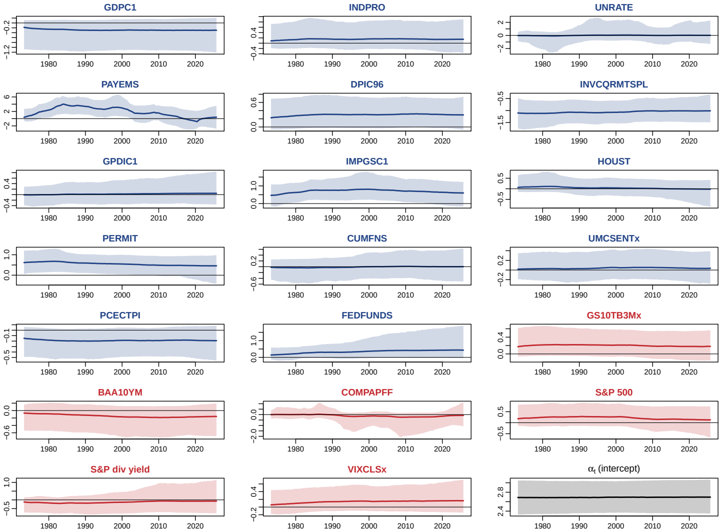
Application: Growth at risk, now TVPs and SV

- **Recall Session 2**: A static Big Data regression of GDP growth on K FRED-QD predictors with a horseshoe prior
- **Now**: Let both the coefficients and the error variance move
- Use a **TVP regression with SV** and the **same horseshoe** prior on both the constant parts and the time-variation scales
- **FRED-QD data**: Target is annualized real GDP growth on a few hand-selected predictors; a full TVP regression with all $K = 231$ series would take ages

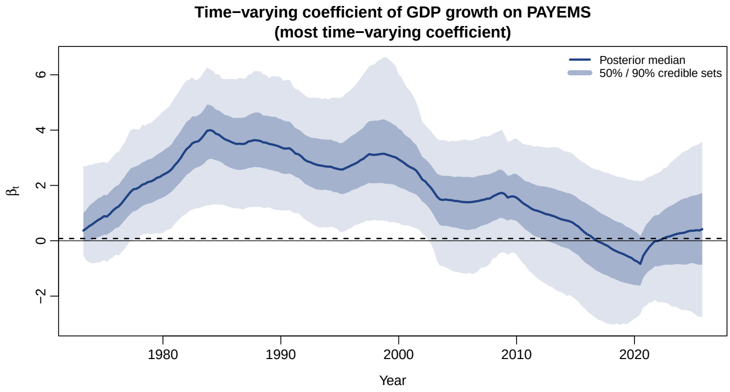
Application: Which coefficients actually vary?



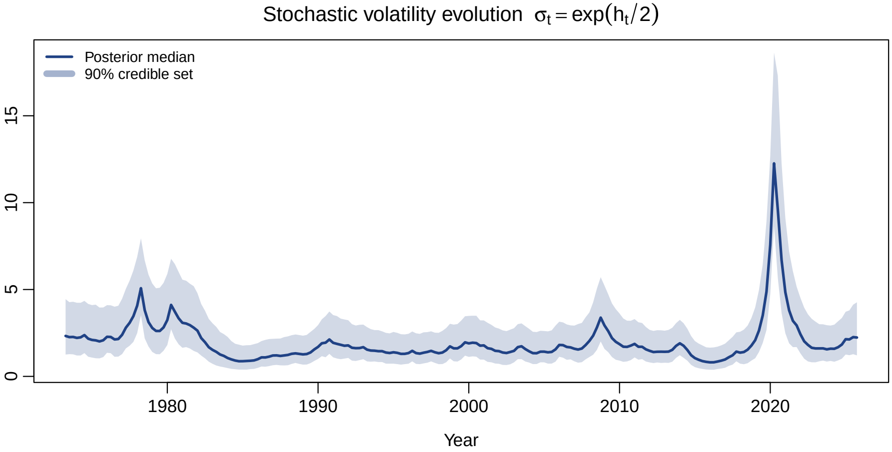
Application: The estimated time-varying coefficients



Application: The most time-varying coefficient



Application: Stochastic volatility in GDP growth



`shrinkTVP`: A great package

- We built the TVP regression with SV **block by block**; the provided codes show how **Bayesian modularity** works
- In applied work, the **shrinkTVP** package does all of this for you; a TVP model is fitted as easily as a linear model with `lm()` in R
- **What it offers out of the box**:
  - A wide range of **shrinkage priors** on both levels and time-variation
  - Built-in **stochastic volatility**
  - Even **dynamic shrinkage** (shrinkage that varies over time)
- The authors develop the **triple gamma** prior, a unifying prior that **nests the horseshoe, and many more** as special cases (Cadonna, Frühwirth-Schnatter and Knaus, 2020, *Econometrics*; Knaus et al., 2021, *JSS*)

A few words on static versus dynamic sparsity

Static sparsity via the non-centered parameterisation

- The NC parameterisation writes the TVP regression as:

$$y_t = \boldsymbol{x}_t' \boldsymbol{\beta}_0 + \sum_{j=1}^{K} x_{jt} \sqrt{\omega_j}\, \tilde{\beta}_{jt} + \varepsilon_t$$

  with the $\sqrt{\omega_j}$s governing the **amount of time variation**

- Shrinking $\sqrt{\omega_j} \to 0$ turns off time variation in β_{jt} entirely; this is **static sparsity**:
  - $\sqrt{\omega_j} = 0$: Coefficient j is **constant for all** t (globally switched off)
  - $\sqrt{\omega_j} > 0$: Coefficient j is **time varying for all** t (globally switched on)

- Cannot capture **occasional** changes, e.g., coefficient that is constant most of the time but shifts during a crisis or a regime change

Dynamic sparsity: occasional time-variation

- **Idea**: let the innovation variance itself move over time, ω_{jt} instead of ω_j:

$$\beta_{j,t+1} = \beta_{jt} + u_{jt}, \quad u_{jt} \sim \mathcal{N}(0, \omega_{jt})$$

  with $\omega_{jt} \approx 0$ locally constant and $\omega_{jt} > 0$ locally varying

- Can **approximate discrete regime changes** without fixing the number of regimes, but requires ω_{jt} to be modelled as a latent process
- **Popular approaches**:
  - **Mixture-innovation models** switch the state innovation on and off: Giordani and Kohn (2008, *JBES*), Nakajima and West (2013, *JBES*), Huber, et al. (2019, *JAE*)
  - **Dynamic global-local shrinkage** priors let the innovation variances follow an SV process: Kowal, Matteson and Ruppert (2019, *JRSS:B*), Hauzenberger, Huber and Koop (2024, *SNDE*)

Key takeaways from Session 3

- UCMs, TVP regressions, SV and DSGEs are all **Gaussian linear state space models** and share the same MCMC machinery (FFBS)
- The **non-centered parameterisation** turns time-variation into a regression problem, so the **Session 2 horseshoe** decides which coefficients actually move
- **Stochastic volatility** adds a nonlinear log-variance state; Kim Shephard Chib algorithm restores tractability

Appendix

Posterior simulation: FFBS, conjugate priors, and stochastic volatility

Posterior simulation: Kalman filtering and smoothing

- For given values of the **system matrices** $\boldsymbol{Z}_t$, $\boldsymbol{T}_t$, $\boldsymbol{\Sigma}_t$, and $\boldsymbol{\Omega}$, posterior simulators for the full state path $\boldsymbol{\beta}^{(1:T)} = (\boldsymbol{\beta}_1', \ldots, \boldsymbol{\beta}_T')'$ exist
- Classic **forward filtering backward sampling (FFBS)** algorithms: Carter and Kohn (1994, *Btka*), Frühwirth-Schnatter (1994, *JTSA*), de Jong and Shephard (1995, *Btka*), Durbin and Koopman (2002, *Btka*)
- These algorithms use **Kalman filtering** (estimating a state at time t using data up to t) and **Kalman smoothing** (estimating a state at time t using all data up to T)
- A modern precision-based sampler is available from Joshua Chan

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Priors for the variances are standard

- It is convenient to use **Inverse-Gamma** priors for the variances σ^2 and ω_j:

$$
\begin{aligned}
\sigma^2 &\sim \mathcal{IG}(\underline{a}_\sigma, \underline{b}_\sigma) \\
\omega_j &\sim \mathcal{IG}(\underline{a}_\omega, \underline{b}_\omega), \quad j = 1, \ldots, K
\end{aligned}
$$

- See Session 2 for the posterior updates of Inverse-Gamma variance parameters
- With stochastic volatility, the constant σ^2 is replaced by $\exp(h_t)$ and the SV block

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A sketch of the MCMC algorithm

- The Gibbs sampler cycles through the following conditional draws:

$$p\left(\sigma^2 \mid \boldsymbol{y}^{(1:T)}, \boldsymbol{\beta}^{(1:T)}, \boldsymbol{\Omega}\right) \quad p\left(\boldsymbol{\Omega} \mid \boldsymbol{y}^{(1:T)}, \sigma^2, \boldsymbol{\beta}^{(1:T)}\right) \quad p\left(\boldsymbol{\beta}^{(1:T)} \mid \boldsymbol{y}^{(1:T)}, \sigma^2, \boldsymbol{\Omega}\right)$$

- For $p(\boldsymbol{\beta}^{(1:T)} \mid \boldsymbol{y}^{(1:T)}, \sigma^2, \boldsymbol{\Omega})$: Use the standard state space algorithm (e.g., Carter and Kohn, 1994).

- For $p(\sigma^2 \mid \bullet)$ and $p(\omega_j \mid \bullet)$: Conditional posteriors have Inverse-Gamma form, exactly as in the natural-conjugate Normal linear regression model

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SV: Properties and initial conditions

- μ_h is the unconditional mean of h_t
- If $|\phi_h| < 1$ (stationary), a natural initial condition is:

$$h_0 \sim \mathcal{N}\left(\mu_h, \frac{\omega_h}{1 - \phi_h^2}\right)$$

- If $\phi_h = 1$, μ_h drops out and one specifies $h_0 \sim \mathcal{N}(\underline{h}, \underline{V}_h)$ directly
- MCMC algorithm draws sequentially from:

$$p(\boldsymbol{h}^{(1:T)} \mid \boldsymbol{y}^{(1:T)}, \mu_h, \phi_h, \omega_h), \quad p(\phi_h \mid \bullet), \quad p(\mu_h \mid \bullet), \quad p(\omega_h \mid \bullet)$$

- The last three conditionals have standard closed forms; drawing $\boldsymbol{h}^{(1:T)}$ requires a dedicated algorithm; see next two slides

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MCMC algorithm for the SV model

- The MCMC algorithm draws sequentially from:

$$p(\boldsymbol{h}^{(1:T)} \mid \boldsymbol{y}^{(1:T)}, \mu_h, \phi_h, \omega_h)$$

$$p(\phi_h \mid \boldsymbol{y}^{(1:T)}, \mu_h, \omega_h, \boldsymbol{h}^{(1:T)}) \quad p(\mu_h \mid \boldsymbol{y}^{(1:T)}, \phi_h, \omega_h, \boldsymbol{h}^{(1:T)}) \quad p(\omega_h \mid \boldsymbol{y}^{(1:T)}, \mu_h, \phi_h, \boldsymbol{h}^{(1:T)})$$

- The last three conditionals have standard closed forms based on results from the normal linear regression model
- Several algorithms exist for drawing $\boldsymbol{h}^{(1:T)}$
- A popular one is from Kim, Shephard and Chib (1998, *ReStud*)

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Mixture of Normals: Component indicators

- The mixture can be written using component indicator variables $s_t \in \{1, 2, \ldots, 7\}$:

$$\varepsilon_t^* \mid s_t = i \sim \mathcal{N}(m_i, v_i^2) \qquad \Pr(s_t = i) = q_i$$

- The MCMC algorithm augments the state space and draws from $p(\boldsymbol{h}^{(1:T)} \mid \boldsymbol{y}^{(1:T)}, \mu_h, \phi_h, \omega_h, \boldsymbol{s}^{(1:T)})$ instead
- Conditional on $\boldsymbol{s}^{(1:T)}$, the problem reduces to a **normal linear** state space model and the standard algorithm applies
- Drawing $p(\boldsymbol{s}^{(1:T)} \mid \boldsymbol{y}^{(1:T)}, \mu_h, \phi_h, \omega_h, \boldsymbol{h}^{(1:T)})$ has a simple closed form; see Kim, Shephard and Chib (1998) for details

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